

Title of the study:

Understanding the Student Experience in the UTSC Computational Physiology Group (CPG)

Faculty investigator:

Dr. Jason Brown, Department of Biological Sciences, University of Toronto Scarborough, Phone 416-287-7475, Email nysuloem.brown@utoronto.ca.

Collaborators:

Terry Yavorsky, former co-lead of CPG and now a graduate student, continuing as a collaborator on this study, University of Toronto (St. George), Email: taras.yavorsky@mail.utoronto.ca

Gursimar Taunk, an undergraduate student conducting this study as his BIOD98 research project under Dr. Brown's academic supervision, University of Toronto Scarborough, Email: gursimar.taunk@mail.utoronto.ca

The purpose of this letter is to help you make an informed decision about whether to consent to participating in this study. This letter will explain the purpose of the study, the possible benefits and risks associated with your participation in this study, and how your survey and interview responses will be kept confidential. If there is any information in this letter that you do not understand, please contact the faculty investigator prior to giving your consent.

The purpose of this study:

The UTSC Computational Physiology Group (CPG) is a group of undergraduate students, led by Dr. Jason Brown, who modify and add to computational models within HumMod, an integrative computational physiology program.

Universities are increasingly looking for models like CPG that can deliver meaningful research experiences without requiring wet-lab space; however, little is currently known about why students choose to join CPG, what they gain from their participation, what challenges they encounter, and whether their experience compares favourably to other research, academic, or community opportunities available to them.

The purpose of this study is: i) to understand students' motivations for joining CPG and the experiences they have while members; ii) to assess whether and how CPG participation benefits students in their academic performance, research skill development, and/or sense of community; iii) to identify challenges members face (e.g., with HumMod, time availability, group dynamics) so that CPG's teaching and onboarding methods can be improved; and iv) to evaluate whether the CPG model could be adapted by other educators at UTSC or other institutions to provide similar undergraduate research experiences elsewhere. Results from this study may be submitted for publication in a scientific/pedagogical research journal and/or presented at academic conferences.

Eligibility for participation in this study:

Anyone who is a current or former member of CPG is eligible to participate in this study.

Procedures of the study:

If you agree to participate, you will complete two components:

- 1) A short (approximately 2-minute) background survey, completed online via a Railway.app application prior to your interview, covering your program enrolment, time at UTSC, time as a CPG member, prior experience, and self-rated confidence in areas such as coding, research, math, and physiology.
- 2) A one-on-one interview, lasting approximately 30–45 minutes, conducted by Gursimar Taunk (see Collaborators, above). Gursimar is not a CPG member. The interview will ask about your motivations for joining CPG, your experiences and challenges as a member, and how CPG compares to other research, academic, or community experiences you may have had. With your permission, the interview will be audio recorded for the purposes of transcription: recordings will be deleted once an accurate transcript has been produced. Interviews will be conducted virtually via Teams, at your convenience.

To participate, you will first complete the background survey—submitting it indicates your willingness to participate in this study. Mr. Taunk will then email you to schedule your interview. Mr. Taunk will link your survey responses to your interview responses and de-identify the combined record once your withdrawal window has closed (see below). You are permitted to skip any questions in the survey or interview that you do not wish to answer, and to decline the audio recording (with notes taken instead), without penalty.

As compensation for your time, you will receive a \$10 gift card for attending your scheduled session, regardless of whether you consent to participate in the interview or to having your responses included in this study.

Benefits of consenting to participate in this study:

Consenting to participate in this study may not provide any direct benefit to you; however, reflecting on your CPG experience may help you better articulate your research and technical skills for graduate/professional school applications or job interviews. More broadly, the results of this study will help Dr. Brown improve CPG's structure, teaching methods, and onboarding process for future members, and may help educators at other institutions develop similar undergraduate research programs.

Risks of consenting to participate in this study:

Consenting to participate in this study carries minimal risk. You may experience minor psychological discomfort when reflecting on your learning experiences, particularly if you found certain aspects of CPG challenging. You may also experience mild anxiety about providing honest or critical feedback about CPG, Dr. Brown, or Mr. Yavorsky. To minimize this risk, interviews are conducted by Gursimar Taunk rather than by Dr. Brown or Mr. Yavorsky, and your individual responses will never be shared with either of them in an identifiable form (see Confidentiality, below). Your decision whether to participate, and anything you say

during the interview, will have no effect on your standing, role, or mentorship relationship within CPG, nor on any current or future course grades.

Confidentiality and anonymity of interview responses:

Because this study involves one-on-one interviews, Gursimar Taunk will necessarily know your identity during the interview itself; however, the faculty investigator (Dr. Brown) and co-investigator (Mr. Yavorsky) will not conduct or be present for interviews, and neither will have access to audio recordings, raw transcripts, or any information that could identify which responses came from which participant.

Mr. Taunk will link your survey responses to your interview transcript, then remove or replace all names and identifying details (e.g., specific project names, dates, or details that could identify you) when preparing the combined, de-identified record. Only this de-identified data will be shared with the faculty investigator for analysis. Audio recordings will be deleted once a transcript has been produced and verified. Individual responses will never be reported; only aggregate, anonymized themes and quotations that cannot be traced back to you will be presented in any publications or presentations.

The de-identified interview data (with no names or identifying information) may be shared with Drs. William Pruett and John Clemmer (University of Mississippi Medical Center), who are developers of HumMod and may be co-authors on any publications resulting from this study. The de-identified data may also be shared with other researchers in the future upon request, provided appropriate data sharing agreements are in place. De-identified data will be stored in encrypted, password-protected form and will only be accessible to the faculty investigator, Mr. Yavorsky, and Mr. Taunk.

Data retention:

Audio recordings will be permanently deleted once transcripts have been produced and verified for accuracy (expected within two weeks of your interview). De-identified survey and transcript data will be retained indefinitely in encrypted form on password-protected computers to support the published research findings and potential future analysis. Only aggregate, anonymized data will be reported in any publications or presentations.

Funding for this study:

No funding has been received for the completion of this study.

Ethics approval:

This study is currently under review by a University of Toronto Research Ethics Board and will be fully compliant with the Tri-Council Policy Statement on Ethical Conduct for Research Involving Humans. This section will be updated with the protocol number once the review is complete.

If you have questions about the ethics of this study, or if you experience any adverse effects as a result of your participation, please contact Mariya Gancheva, Research Ethics Coordinator, Human Research Ethics, Research Oversight & Compliance Office, (416) 946-3273, m.gancheva@utoronto.ca.



UNIVERSITY OF
TORONTO
SCARBOROUGH

Your decision to consent to participate in this study is completely voluntary:

Because this study involves two stages, you will be asked to confirm your consent twice. Before completing the background survey, you will confirm that you have read this Information and Consent Letter and agree to have your background information collected for this study. Before your interview begins, you will confirm your consent again. This second confirmation is what indicates your agreement to participate in the interview and to have your combined survey and interview responses included in the study. You may decline consent at either point, or withdraw partway through the interview, and this will have no effect on your standing or role within CPG. If you consent and complete the interview, you may still withdraw your data for up to two weeks following your interview. Mr. Taunk will email you 48 hours before this deadline as a reminder of your option to withdraw. After this period, your questionnaire and interview data will be de-identified and your specific responses can no longer be located and removed, because they are no longer linked to your name.

Additional information:

If you have any questions regarding this study, or would like additional information to assist you in deciding whether to give consent to participate, please contact the faculty investigator listed above.